FLOREA DARIA-MIHAELA
311CC

Examen final

3. $f(x) = \begin{cases} k \arctan x, & x \in [0, \frac{\pi}{4}] \\ 0, & \text{in rest} \end{cases}$

a) k constantă a.î. f = densitate de probabilitate

Punem condiția:

$$\int_{-\infty}^{+\infty} f(x)\,dx = 1 \iff \int_{0}^{\pi/4} k \arctan x\,dx = 1 \iff$$

$$k \cdot \int_{0}^{\pi/4} \arctan x\,dx = 1 \iff k \cdot \left(x\arctan x \Big|_0^{\pi/4} - \int_0^{\pi/4} \frac{x}{x^2+1}\,dx \right) = 1 \iff$$

$$k \cdot \frac{1}{2}\left(\frac{\pi}{2}\arctan\left(\frac{\pi}{4}\right) - \ln\left(\frac{\pi^2}{16}+1\right)\right) = 1 \iff k = \frac{2}{\frac{\pi}{2}\arctan\frac{\pi}{4} - \ln\left(\frac{\pi^2}{16}+1\right)}$$

Cum $K > 0 \Rightarrow f(x) \geq 0$

Deci f este densitate de probabilitate

pt. $k = \dfrac{2}{\frac{\pi}{2}\arctan\frac{\pi}{4} - \ln\left(\frac{\pi^2}{16}+1\right)}$

b) Funcția de repartiție

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt = k \cdot \int_{0}^{x} \arctan t\,dt =$$

$$= k \cdot \left(x \arctan x - \frac{\ln(x^2+1)}{2} \right), \text{ pentru } x \in [0, \frac{\pi}{4}]$$

Pentru $x > \frac{\pi}{4}$ $\Rightarrow$

$$F_X(x) = \int_{0}^{\pi/4} f(t)\,dt = 1 \quad \text{(conform subpunctului a))}$$

$$\Rightarrow F_X(x) = \begin{cases} 0, & \text{pentru } x < 0 \\ k \cdot \left(x \arctan x - \frac{\ln(x^2+1)}{2} \right), & x \in [0, \frac{\pi}{4}] \\ 1, & x > \frac{\pi}{4} \end{cases}$$

c) $$E(X) = \int_{-\infty}^{+\infty} x f(x)\,dx = k \cdot \int_{0}^{\pi/4} x \cdot \arctan x\,dx$$

$$= k \cdot \frac{1}{2} \left[(x^2+1) \arctan x - x \right] \Big|_0^{\pi/4} =$$

$$= k \cdot \frac{(\pi^2+16) \arctan\left(\frac{\pi}{4}\right) - 4\pi}{32}$$

(4.) $X \sim N(0,1)$

$Y \sim N(0,2)$

$P((x,y) \in D)$

$D = \{(x,y) \in \mathbb{R} \mid x^2 + \frac{y^2}{4} \leq 1\}$

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \cdot e^{\frac{-x^2}{2}}, \quad x \in \mathbb{R}$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \cdot e^{\frac{-y^2}{8}} \quad , y \in \mathbb{R}$$

Cum X și Y sunt independente $\Rightarrow$

$$f_{(X,Y)}(x,y) = f_X(x) \cdot f_Y(y) =$$

$$= \frac{1}{4\pi} \cdot e^{-\left(\frac{x^2}{2} + \frac{y^2}{8}\right)}$$

$$P((X,Y) \in D) = \iint_D f(x,y)\, dx\, dy$$

$$D = \{ (x,y) \in \mathbb{R} \mid x^2 + \frac{y^2}{4} \leq 1 \}$$

Parametrizare

$$\begin{cases} x = \rho \cos\theta \\ y = 2\rho \sin\theta \end{cases}$$

$\rho^2 \leq 1 \Rightarrow \rho \in [0,1]$

$\theta \in [0, 2\pi]$

Jacobian 2ρ

$$\Rightarrow \iint_D f(x,y)\, dx\, dy = \int_0^{2\pi} \int_0^1 \frac{1}{4\pi} e^{-\frac{1}{2}\cdot\rho^2} \cdot 2\rho\, d\rho\, d\theta$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \int_0^1 e^{-\frac{\rho^2}{2}} \cdot \rho\, d\rho\, d\theta =$$

$$= \frac{1}{2\pi} \cdot \int_0^{2\pi} -e^{-\frac{\rho^2}{2}} \Big|_0^1 d\theta =$$

$$= \frac{1}{2\pi} \int_0^{2\pi} (-e^{-\frac{1}{2}} + 1)\, d\theta = (1 - e^{-\frac{1}{2}}) \cdot \frac{1}{2\pi} \cdot 2\pi =$$

$= 1 - e^{-\frac{1}{2}}$

2. $\begin{cases} 3\ F_1 \rightarrow p_1 = 0{,}9 \\ 5\ F_2 \rightarrow p_2 = 0{,}75 \\ 2\ F_3 \rightarrow p_3 = 0{,}85 \end{cases}$

a) Vrem folosi formula probabilității totale:

A = evenimentul „aparatul trece proba de verificare"

B_i = evenimentul „este luat aparat de la furnizorul F_i"

$P(A) = P(A|B_1) \cdot P(B_1) + P(A|B_2) \cdot P(B_2) + P(A|B_3) \cdot P(B_3)$

$P(B_1) = 0{,}3 \qquad P(A|B_1) = 0{,}9$

$P(B_2) = 0{,}5 \qquad P(A|B_2) = 0{,}75$

$P(B_3) = 0{,}2 \qquad P(A|B_3) = 0{,}85$

$\Rightarrow P(A) = 0{,}9 \cdot 0{,}3 + 0{,}75 \cdot 0{,}5 + 0{,}85 \cdot 0{,}2 =$

$= 0{,}815$

b) Probabilitatea cerută este $P(B_1|A)$

Vrem folosi formula lui Bayes

$$P(B_1/A) = \frac{P(A/B_1)\cdot P(B_1)}{\sum_{i=1}^{3} P(A/B_i)\cdot P(B_i)} =$$

$$= \frac{0,9 \cdot 0,3}{P(A)} = \frac{0,9 \cdot 0,3}{0,815} = 0,331$$

(5.) $p = 0,75 \Rightarrow q = 0,25$

Conform teoremei Moivre-Laplace

avem $P\left(\sqrt{\frac{n}{pq}}\,|X_n - p| \leq \beta\right) \approx 2\Phi(\beta) - 1$

Cunoaștem că $2\Phi(\beta) - 1 = 0,96 \Rightarrow$

$\Phi(\beta) = 0,98 \Rightarrow \beta = 2,05$

Dar $\beta \leq 2,05 = \sqrt{\frac{n}{pq}} \cdot 0,05 = \sqrt{\frac{n}{0,25 \cdot 0,75}} \cdot 0,05 \Rightarrow$

$\sqrt{n} = 17,753 \Rightarrow n = 315,169$

Dar $n \in \mathbb{N} \Rightarrow \boxed{n \geq 315}$ de repetări

(1.) 10^9 biți / secundă

$p = 10^{-9}$

Fie X v.a. ce desemnează numărul de erori într-o secundă. $\Rightarrow$

X este poissoniană cu $\lambda = 10^{-9}$ $\Rightarrow$

$$P(X>5) = 1 - P(X\leq 5) = 1 - P(X=0) - P(X=1) - P(X=2) - P(X=3) - P(X=4) - P(X=5)$$

$$P(X=k) = \frac{\lambda^k}{k!} \cdot e^{-\lambda} \quad k \in \mathbb{N}$$

$$P(X=0) = e^{-10^9}$$

$$P(X=1) = 10^9 \cdot e^{-10^9}$$

$$P(X=2) = \frac{10^{18}}{2} \cdot e^{-10^9}$$

$$P(X=3) = \frac{10^{27}}{6} \cdot e^{-10^9}$$

$$P(X=4) = \frac{10^{36}}{24} \cdot e^{-10^9}$$

$$P(X=5) = \frac{10^{45}}{120} \cdot e^{-10^9}$$

$$\Rightarrow P(X>5) = 1 - e^{-10^9} \cdot \left(1 + 10^9 + \frac{10^{18}}{2} + \frac{10^{27}}{6} + \frac{10^{36}}{24} + \frac{10^{45}}{120}\right)$$

6. $n=35$ volmetre

Vom verifica ipotezele statistice parametrice